

Newman-Penrose Formalism Calculations

Stephani et al., Equation (23.40)

Metric

$$ds^2 = -\frac{1}{S^2}dt \otimes dt + \frac{1}{x^2 S^2}dx \otimes dx + \left(\frac{4x^{b^2+b} \cosh(t)^{-b+1}}{S^2} \right) dy \otimes dy + \left(\frac{4x^{b^2-b} \cosh(t)^{b+1}}{S^2} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$S = s_0 \sinh(t)^k + x^k$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}}{2S} \right) dt + \left(\frac{\sqrt{2}}{2xS} \right) dx$$

$$n = \left(\frac{\sqrt{2}}{2S} \right) dt + \left(-\frac{\sqrt{2}}{2xS} \right) dx$$

$$m = \left(\frac{\sqrt{2}x^{\frac{1}{2}b^2}x^{\frac{1}{2}b}\cosh(t)^{-\frac{1}{2}b}\sqrt{\cosh(t)}}{S} \right) dy + \left(\frac{i\sqrt{2}x^{\frac{1}{2}b^2}\cosh(t)^{\frac{1}{2}b}\sqrt{\cosh(t)}}{x^{\frac{1}{2}b}S} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}x^{\frac{1}{2}b^2}x^{\frac{1}{2}b}\cosh(t)^{-\frac{1}{2}b}\sqrt{\cosh(t)}}{S} \right) dy + \left(-\frac{i\sqrt{2}x^{\frac{1}{2}b^2}\cosh(t)^{\frac{1}{2}b}\sqrt{\cosh(t)}}{x^{\frac{1}{2}b}S} \right) dz$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} S \right) dt + \left(\frac{1}{2} \sqrt{2} x S \right) dx$$

$$n = \left(-\frac{1}{2} \sqrt{2} S \right) dt + \left(-\frac{1}{2} \sqrt{2} x S \right) dx$$

$$m = \left(\frac{\sqrt{2} \cosh(t)^{\frac{1}{2}b} S}{4 x^{\frac{1}{2}b^2} x^{\frac{1}{2}b} \sqrt{\cosh(t)}} \right) dy + \left(\frac{i \sqrt{2} x^{\frac{1}{2}b} \cosh(t)^{-\frac{1}{2}b} S}{4 x^{\frac{1}{2}b^2} \sqrt{\cosh(t)}} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2} \cosh(t)^{\frac{1}{2}b} S}{4 x^{\frac{1}{2}b^2} x^{\frac{1}{2}b} \sqrt{\cosh(t)}} \right) dy + \left(-\frac{i \sqrt{2} x^{\frac{1}{2}b} \cosh(t)^{-\frac{1}{2}b} S}{4 x^{\frac{1}{2}b^2} \sqrt{\cosh(t)}} \right) dz$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} S \partial_t + \frac{1}{2} \sqrt{2} x S \partial_x$$

$$\Delta = -\frac{1}{2} \sqrt{2} S \partial_t - \frac{1}{2} \sqrt{2} x S \partial_x$$

$$\delta = \frac{\sqrt{2} \cosh(t)^{\frac{1}{2}b} S}{4 x^{\frac{1}{2}b^2} x^{\frac{1}{2}b} \sqrt{\cosh(t)}} \partial_y + \frac{i \sqrt{2} x^{\frac{1}{2}b} \cosh(t)^{-\frac{1}{2}b} S}{4 x^{\frac{1}{2}b^2} \sqrt{\cosh(t)}} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2} \cosh(t)^{\frac{1}{2}b} S}{4 x^{\frac{1}{2}b^2} x^{\frac{1}{2}b} \sqrt{\cosh(t)}} \partial_y - \frac{i \sqrt{2} x^{\frac{1}{2}b} \cosh(t)^{-\frac{1}{2}b} S}{4 x^{\frac{1}{2}b^2} \sqrt{\cosh(t)}} \partial_z$$

Spin Coefficients

$$\rho = -\frac{1}{4}\sqrt{2}b^2S + \frac{1}{2}\sqrt{2}x\frac{\partial}{\partial x}S + \frac{\sqrt{2}S\sinh(t)}{4\cosh(t)} - \frac{1}{2}\sqrt{2}\frac{\partial}{\partial t}S$$

$$\sigma = -\frac{1}{4}\sqrt{2}bS - \frac{\sqrt{2}bS\sinh(t)}{4\cosh(t)}$$

$$\kappa = 0$$

$$\mu = -\frac{1}{4}\sqrt{2}b^2S + \frac{1}{2}\sqrt{2}x\frac{\partial}{\partial x}S - \frac{\sqrt{2}S\sinh(t)}{4\cosh(t)} + \frac{1}{2}\sqrt{2}\frac{\partial}{\partial t}S$$

$$\lambda = -\frac{1}{4}\sqrt{2}bS + \frac{\sqrt{2}bS\sinh(t)}{4\cosh(t)}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{1}{4}\sqrt{2}x\frac{\partial}{\partial x}S + \frac{1}{4}\sqrt{2}\frac{\partial}{\partial t}S$$

$$\gamma = -\frac{1}{4}\sqrt{2}x\frac{\partial}{\partial x}S - \frac{1}{4}\sqrt{2}\frac{\partial}{\partial t}S$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{1}{8}b^4S^2 - \frac{1}{4}b^2S^2 + \frac{1}{2}x^2S\frac{\partial^2}{(\partial x)^2}S - xS\frac{\partial^2}{\partial t\partial x}S + \frac{1}{2}xS\frac{\partial}{\partial x}S - \frac{1}{8}S^2 + \frac{b^2S^2}{8\cosh(t)^2} + \frac{1}{2}S\frac{\partial^2}{(\partial t)^2}S - \frac{S^2}{8\cosh(t)^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{1}{4}b^3S^2 + \frac{1}{2}bxS\frac{\partial}{\partial x}S - \frac{1}{4}bS^2 + \frac{bS\sinh(t)\frac{\partial}{\partial t}S}{2\cosh(t)}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{1}{16}b^4S^2 + \frac{1}{4}b^2xS\frac{\partial}{\partial x}S - \frac{1}{4}x^2S\frac{\partial^2}{(\partial x)^2}S - \frac{1}{4}xS\frac{\partial}{\partial x}S + \frac{1}{16}S^2 + \frac{b^2S^2}{16\cosh(t)^2} - \frac{S\sinh(t)\frac{\partial}{\partial t}S}{4\cosh(t)} + \frac{1}{4}S\frac{\partial^2}{(\partial t)^2}S - \frac{S^2}{16\cosh(t)^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{1}{4}b^3S^2 + \frac{1}{2}bxS\frac{\partial}{\partial x}S - \frac{1}{4}bS^2 + \frac{bS\sinh(t)\frac{\partial}{\partial t}S}{2\cosh(t)}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{1}{8}b^4S^2 - \frac{1}{4}b^2S^2 + \frac{1}{2}x^2S\frac{\partial^2}{(\partial x)^2}S + xS\frac{\partial^2}{\partial t\partial x}S + \frac{1}{2}xS\frac{\partial}{\partial x}S - \frac{1}{8}S^2 + \frac{b^2S^2}{8\cosh(t)^2} + \frac{1}{2}S\frac{\partial^2}{(\partial t)^2}S - \frac{S^2}{8\cosh(t)^2}$$

$$\Lambda = -\frac{1}{16}b^4S^2 + \frac{1}{4}b^2xS\frac{\partial}{\partial x}S - \frac{1}{2}x^2\frac{\partial}{\partial x}S^2 + \frac{1}{4}x^2S\frac{\partial^2}{(\partial x)^2}S + \frac{1}{4}xS\frac{\partial}{\partial x}S + \frac{1}{16}S^2 - \frac{b^2S^2}{48\cosh(t)^2} - \frac{S\sinh(t)\frac{\partial}{\partial t}S}{4\cosh(t)} + \frac{1}{2}\frac{\partial}{\partial t}S^2 - \frac{1}{4}S\frac{\partial^2}{(\partial t)^2}S + \frac{S^2}{48\cosh(t)^2}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{1}{4}b^3S^2 - \frac{b^3S^2 \sinh(t)}{4 \cosh(t)} + \frac{1}{4}bS^2 + \frac{bS^2 \sinh(t)}{4 \cosh(t)}$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{b^2S^2}{12 \cosh(t)^2} + \frac{S^2}{12 \cosh(t)^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{1}{4}b^3S^2 + \frac{b^3S^2 \sinh(t)}{4 \cosh(t)} + \frac{1}{4}bS^2 - \frac{bS^2 \sinh(t)}{4 \cosh(t)}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"